

Question:

At birth, the least developed part of the brain is the

Answer Choices:

- A. visual system
- B. thalamus
- C. frontal lobe
- D. brain stem
- E. parietal lobe
- F. cerebellum
- G. cortex
- H. basal ganglia
- I. hypothalamus
- J. limbic system

Reasoning + Answer:

</tool_call> reasoning process here

At birth , the brain undergoes rapid development , with certain regions developing earlier than others . The brain stem , which controls basic life-sustaining functions like breathing and heart rate , is one of the first structures to develop and is largely present at birth . The visual system , thalamus , hypothalamus , limbic system , basal ganglia , and cerebellum develop later , with the cerebral cortex and especially the frontal lobe being the last to mature .

Among the options , the least developed part of the brain at birth is the cerebral cortex , particularly the frontal lobe , which is responsible for higher cognitive functions and is the most developed later in life . The cortex is not fully myelinated or organized at birth and undergoes significant development postnatally .

Thus , the least developed part at birth is the cortex , specifically the frontal lobe , which is not mature in structure or function .

- A . visual system â relatively developed at birth , though maturation continues
- B . thalamus â developed early
- C . frontal lobe â least developed at birth , develops over months and years
- D . brain stem â developed early
- E . parietal lobe â develops postnatally
- F . cerebellum â develops steadily over time , mature later
- G . cortex â includes frontal lobe ; least developed at birth
- H . basal ganglia â develops gradually
- I . hypothalamus â present at birth but functional maturation is delayed
- J . limbic system â develops progressively

Among the options , the frontal lobe (part of the cortex) is the most prominent in terms of maturity delay , making it the least developed at birth .

So the best choice from the list that reflects the most underdeveloped region at birth is the frontal lobe (which is part of the cerebral cortex).

However , " cortex " as a whole is widely recognized as developing last . But the question asks for the " least developed part " , and the frontal lobe is the last region to mature , critically for executive functions .

So the answer is either G (cortex) or C (frontal lobe) . However , the " cortex " as a structure is broad , and within it , the frontal cortex is especially underdeveloped at birth . Even so , note that the cortex is the outer layer , and although it develops last , the cortex as a whole is less formed at birth .

But , critical evidence in pediatric neurodevelopment shows the frontal lobe is the last brain region to develop and its maturation takes until early adolescence .

Among the options , both G and C are present , and the frontal lobe is a region of cortex .

But the option C â " frontal lobe " â is specifically known to be the most immature at birth , especially for prefrontal functions .

Final decision : There is some overlap , but the newest neurodevelopmental research shows the frontal lobe , not the whole cortex , is the most underdeveloped . Also , the cortex is structural term ; the frontal lobe is a region ; and the frontal lobe is the actual structure maturing last .

Therefore , the " least developed part " is the frontal lobe .

< answer > C < / answer >